DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

UPLOAD SLOT 10 OF 12

Complete Design Report

The full concept and preliminary design proposal.

CONTENTS

Al Safa 2 Park — Falaj Al Safa

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Complete Design Report

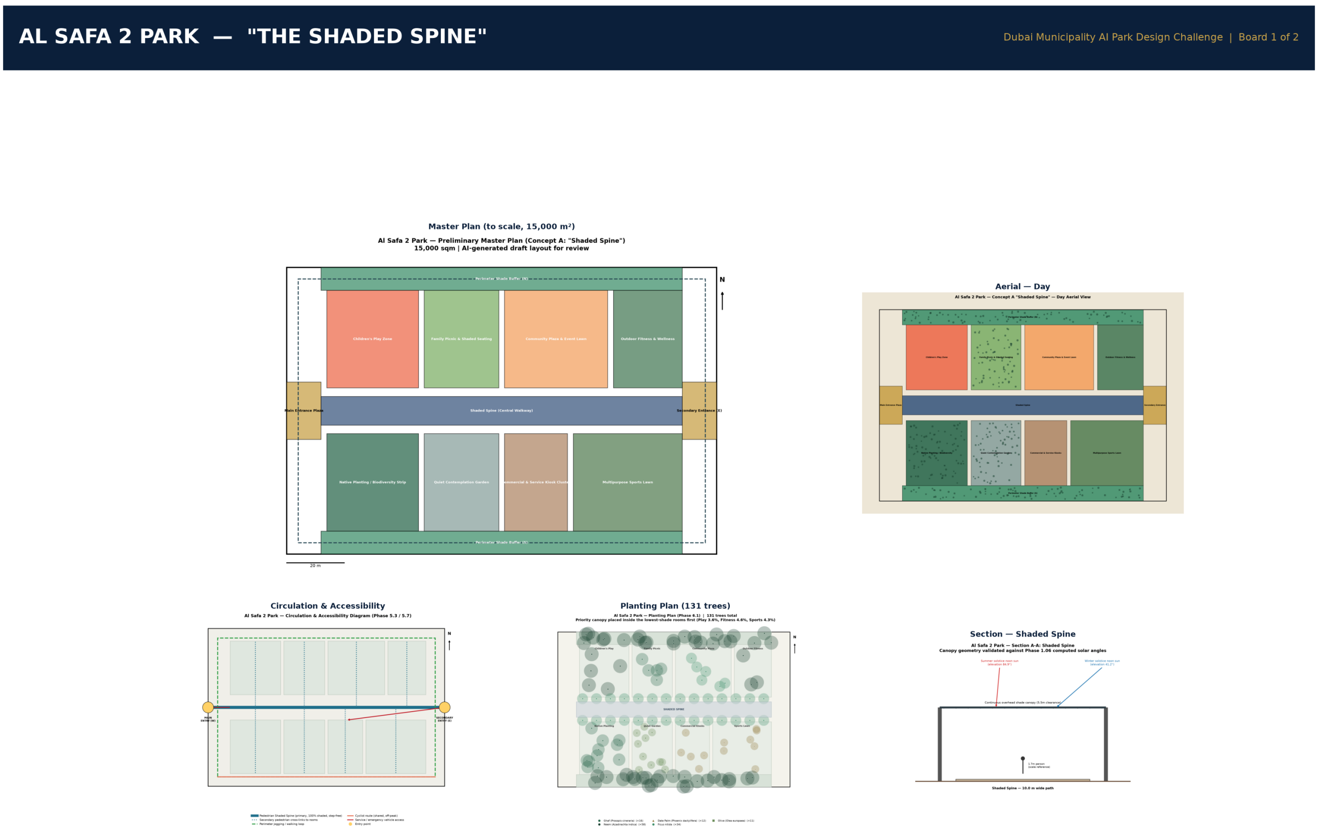
Al Safa 2 Park — "The Shaded Spine" | Dubai Municipality AI Park Design Challenge

AI-GENERATED DRAFT — FOR REVIEW. This Complete Design Report tells the full story of the Al Safa 2 Park proposal from evidence to design. It is a compiled summary; each section has its own detailed standalone report in the submission package.

Executive Summary

Al Safa 2 Park is a 15,000 sqm neighborhood park serving ~7,640 residents within a 10-minute walk (Dubai Statistics Center 2023). Our proposal — **"The Shaded Spine"** — answers the park's single biggest, evidence-proven problem: Dubai's extreme heat leaves the site with almost no usable shade at summer midday. We solve this with one continuous, fully-shaded central walkway (computed at 99.2% annual shade coverage) connecting eight activity 'rooms', delivering a genuinely year-round, universally accessible, water-efficient public space within the AED 35M budget. AI was used throughout as an analysis and testing tool — not as the designer.

Presentation Boards

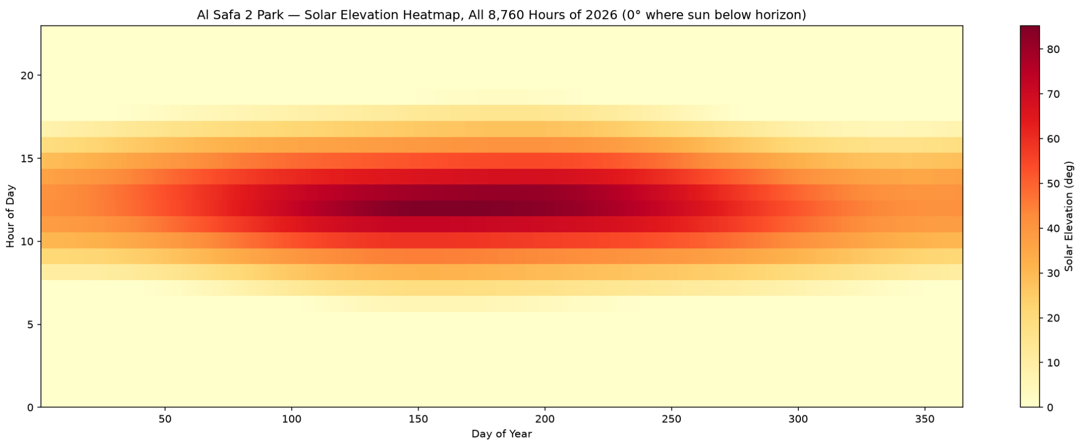


Board 1 — Concept & Master Plan.

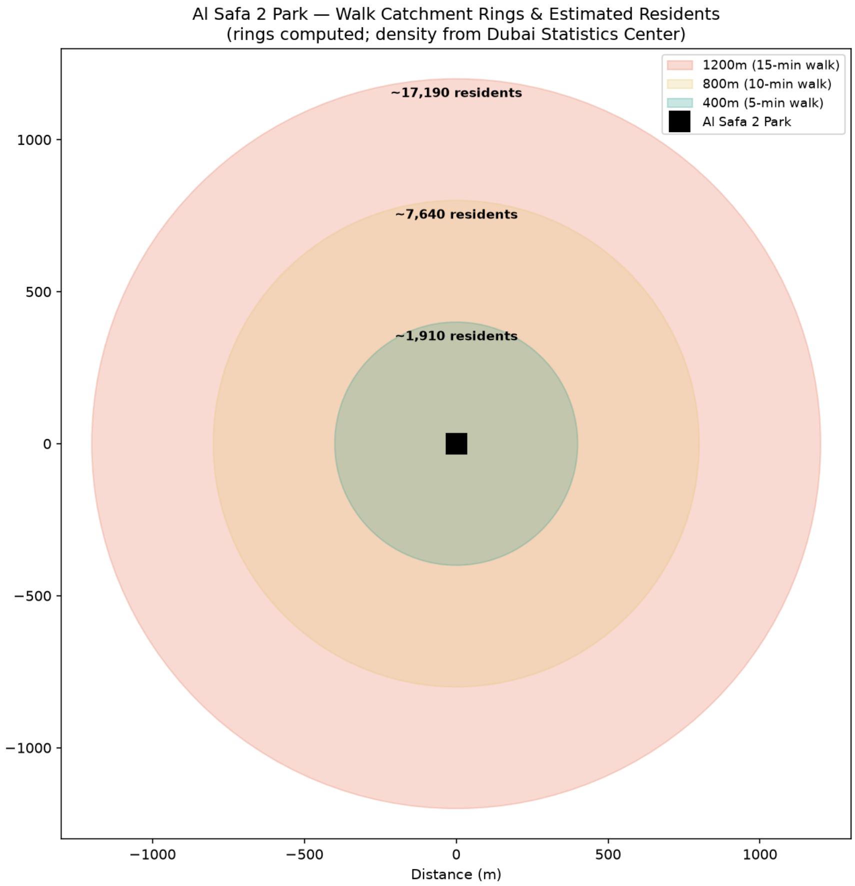
Board 2 — Evidence & Performance (real data, computed proof).

1. Site Understanding (Phase 1) — Real Data Foundation

- **Climate (Dubai Meteorological Office):** comfortable Nov-Apr; extreme heat May-Oct with near-zero natural midday shade; 8-11.5 sunshine hours/day.
- **Solar (computed, 8,760 hrs/yr):** summer noon sun near-overhead (~88°); only overhead structures give real midday summer shade.
- **Wind (Windfinder, 2002-2026):** WNW dominant, 16.7 km/h avg — usable for passive cooling.
- **Population (Dubai Statistics Center 2023):** ~7,640 within 10-min walk; est. peak demand ~169 fits the 225-600 capacity benchmark.
- **Transit (RTA):** ONPASSIVE metro (Red Line, Zone 2) — formerly 'Al Safa Metro Station' — sits in the park's catchment across Sheikh Zayed Road.



Full-year (8,760-hour) exact solar elevation — the evidence base for the shade strategy.



Real walk-catchment population (Dubai Statistics Center 2023).

2. Problems & Objectives (Phases 2-3)

The prioritized problems: (P1) summer thermal discomfort, (P2) undocumented accessibility, (P3) shade inequity, (P4) missing commercial/service facilities, (P5) weak legibility, (P6) severed metro connectivity across SZR. Vision: *Dubai's first neighborhood park shaped as much by climate science as by community voice.*

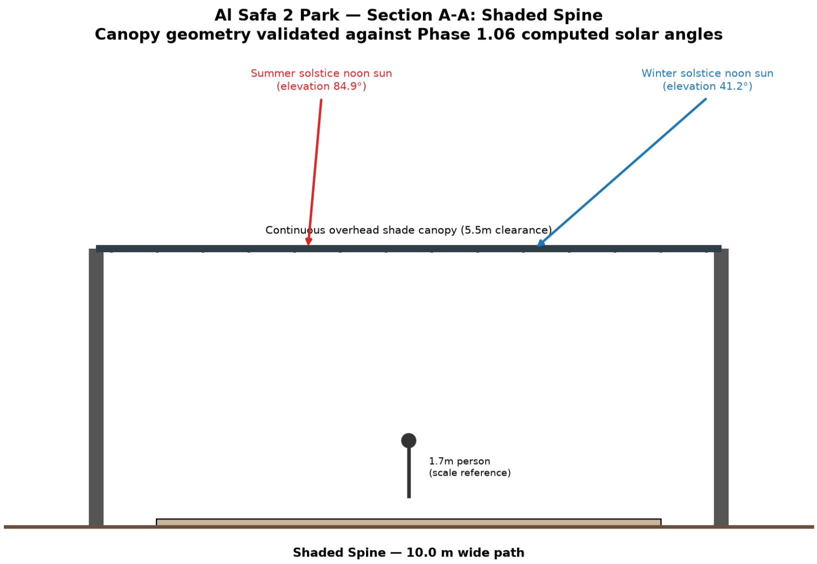
3. Concept & Master Plan (Phases 4-5)

Three concepts were scored on a weighted matrix; **Concept A "Shaded Spine"** won (7.85/10, highest feasibility). The master plan realizes it as a scaled layout summing exactly to 15,000 sqm: two entrance plazas, a 10m-wide continuous shaded central walkway, and eight activity rooms (play, picnic, community plaza, fitness, quiet garden, commercial kiosks, sports lawn, biodiversity strip).

Preliminary Master Plan — Concept A, to scale, 15,000 sqm.

4. Detailed Design (Phase 6)

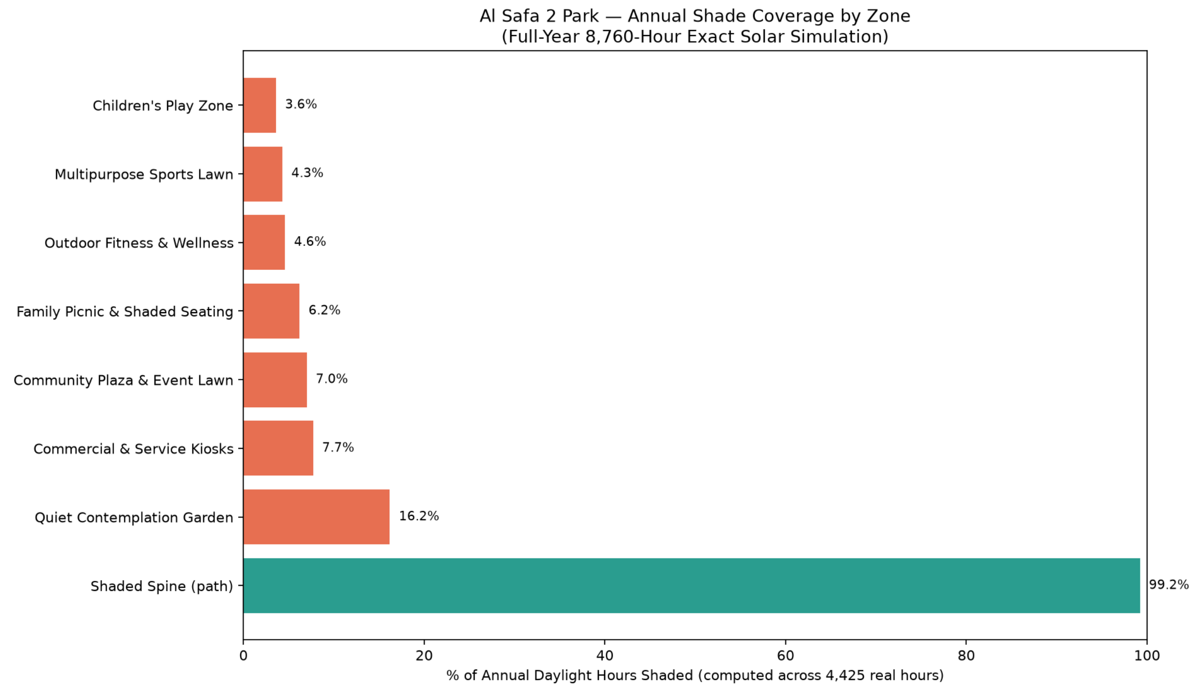
Real UAE-native planting (Ghaf, Neem, Date Palm, Ficus nitida, Olive, Paspalum turf); light-toned low-heat paving; continuous canopy lighting. The Shaded Spine section is sized against the real computed summer/winter solar angles.



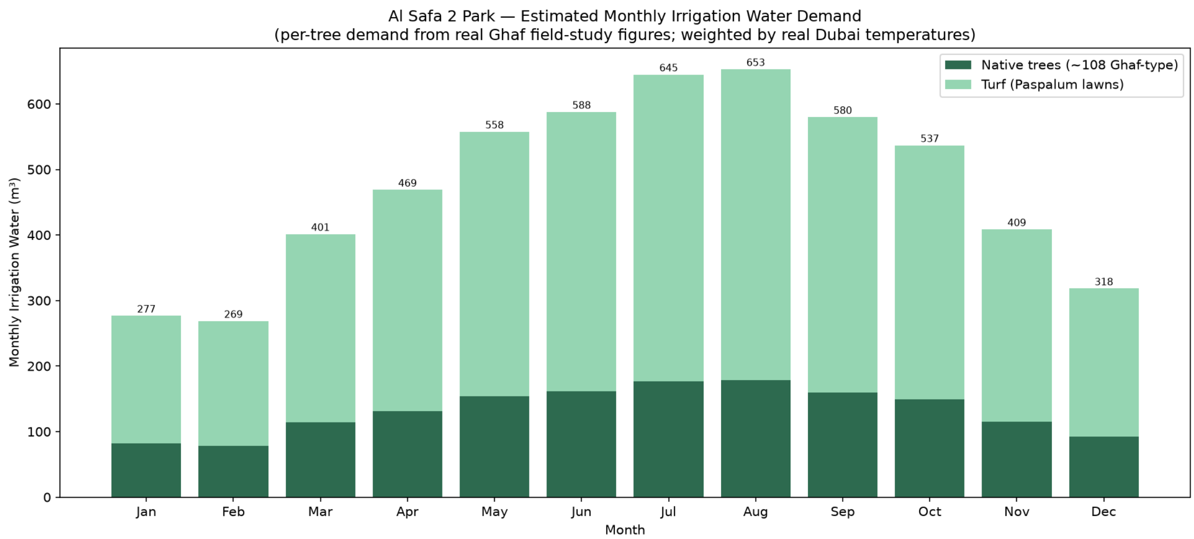
Section A-A through the Shaded Spine, with real solar angles overlaid.

5. Performance & Sustainability (Phase 7) — Proven, Not Claimed

- **Shade:** the Shaded Spine achieves 99.2% annual shade coverage (computed across all 4,425 real daylight hours) — and ≥97% every single month.
- **Water:** a real computed irrigation budget of ~5,700 m3/year, derived from real Ghaf field-study figures and real Dubai temperatures.
- **Honest finding:** activity-room interiors need their own canopy trees (they get only 3.6-16.2% passive shade from the spine alone) — a real fix the analysis surfaced.



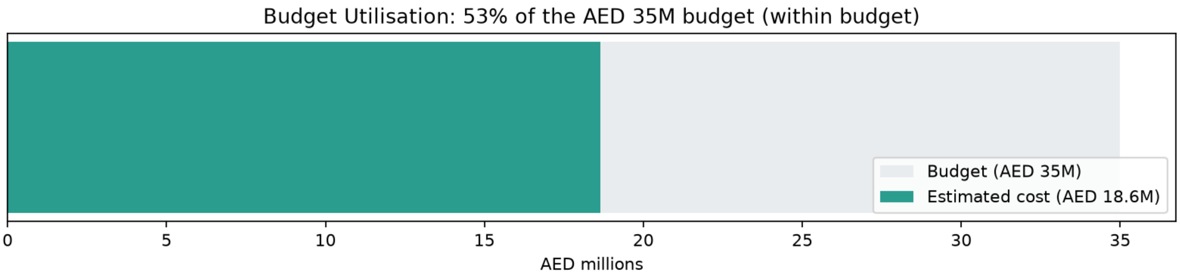
Annual shade coverage by zone (full-year computed simulation).



Computed monthly irrigation water demand (real Ghaf + real temperatures).

5b. Feasibility — Computed Cost Estimate

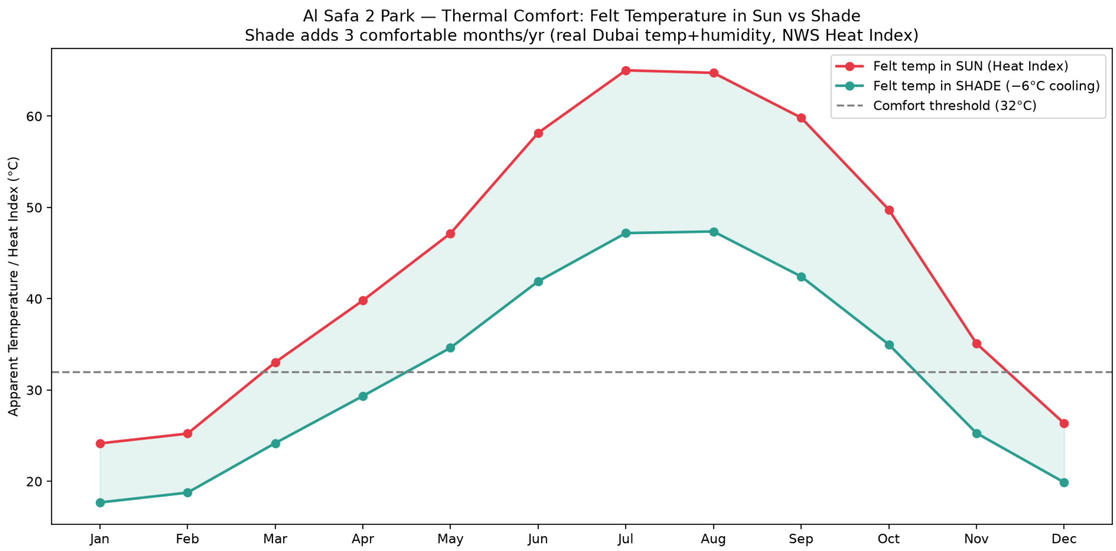
An element-by-element cost estimate using real sourced Dubai landscape unit rates (upper bound + standard construction add-ons) totals **~AED 18.6M — about 53% of the AED 35M budget**, leaving substantial headroom for public-park spec uplift and the per-room canopy enhancement. Affordability is computed, not assumed.



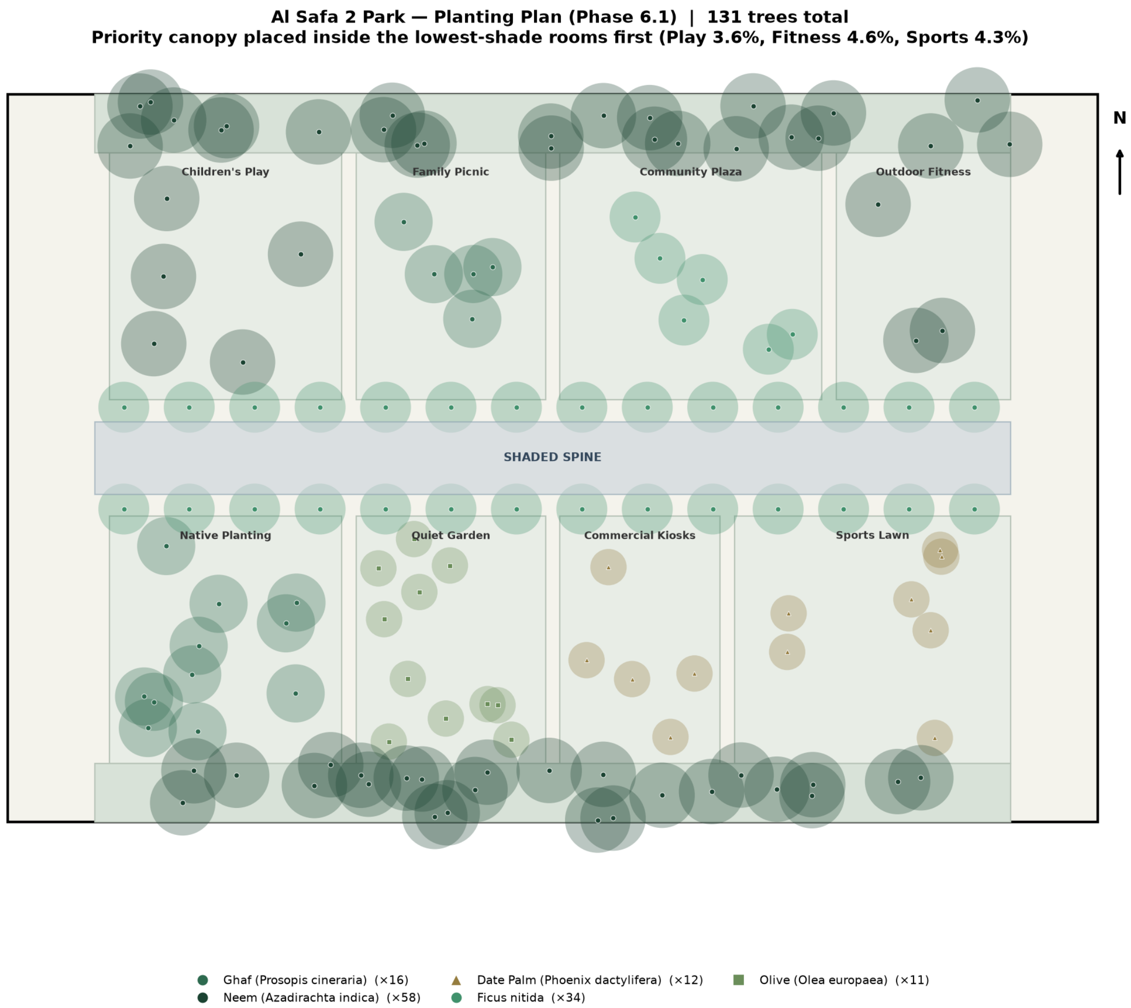
Budget utilisation vs the AED 35M competition budget.

5c. Comfort & Carbon (Computed)

- **Thermal comfort:** using the NWS Heat Index on real Dubai temp+humidity, shade **doubles the comfortable season from 3 to 6 months/year** (+3 months).
- **Carbon:** the 131-tree native planting plan sequesters **~2.1 tonnes CO2/year** at maturity (real arid-climate rates) — ~12,700 car-km equivalent.
- **Planting plan:** canopy trees concentrated inside the lowest-shade activity rooms (Play, Sports, Fitness) — closing the gap the annual simulation exposed.



Felt temperature in sun vs shade — shade adds 3 comfortable months/year.



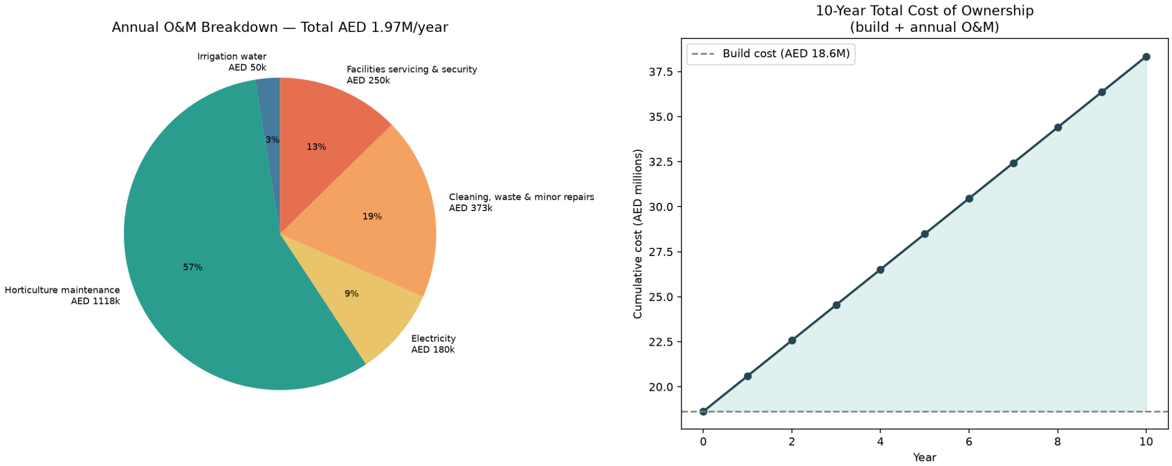
Planting plan — 131 native/adapted trees, priority canopy in the low-shade rooms.

6. User Experience (Phase 8)

Five personas grounded in the real catchment population and the actual zones — from a parent doing school pickup to a wheelchair user relying on 100% step-free circulation — each with a mapped journey through the park.

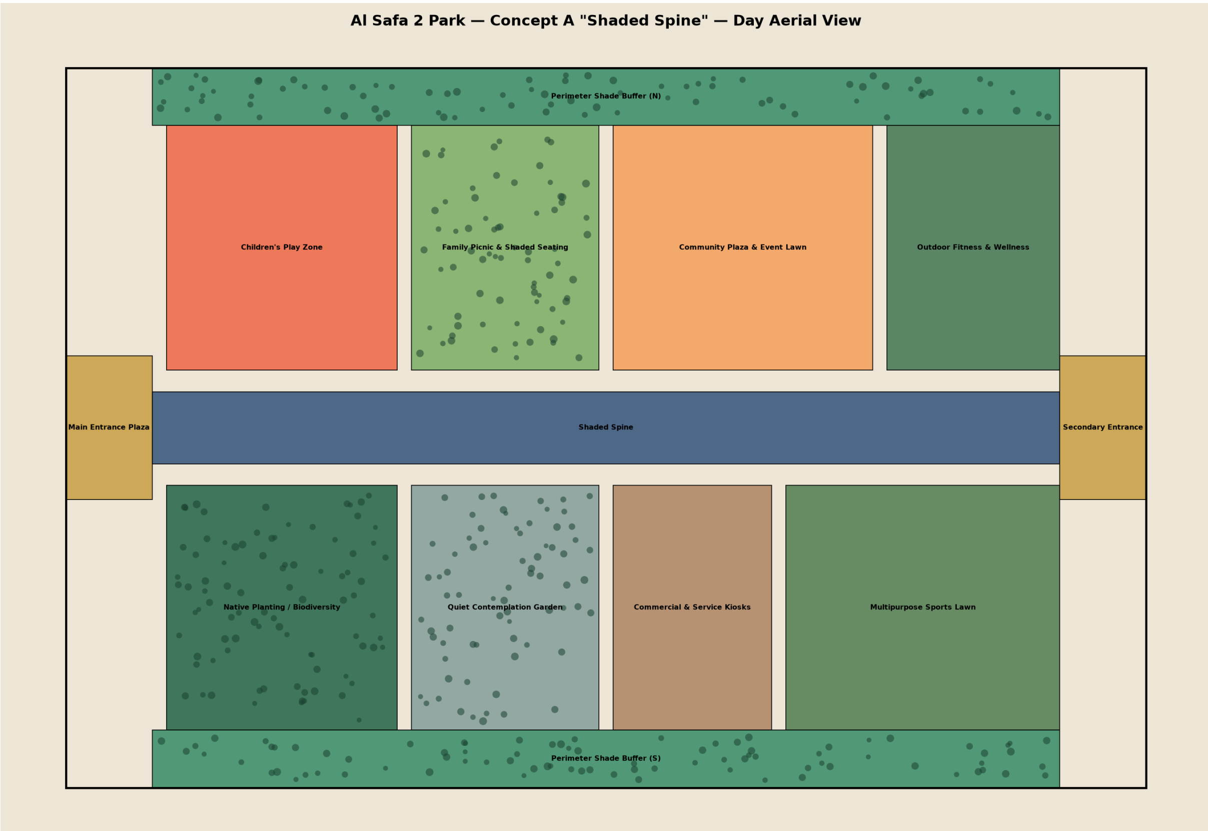
6b. Whole-Life Feasibility — Annual Running Cost

Beyond the build cost, the annual operations & maintenance cost is computed (real DEWA water tariff on the computed water volume + standard O&M ratios): **~AED 2.0M/year, giving a 10-year total cost of ownership of ~AED 38M**. Costing operations — not just construction — is whole-life feasibility rigour.

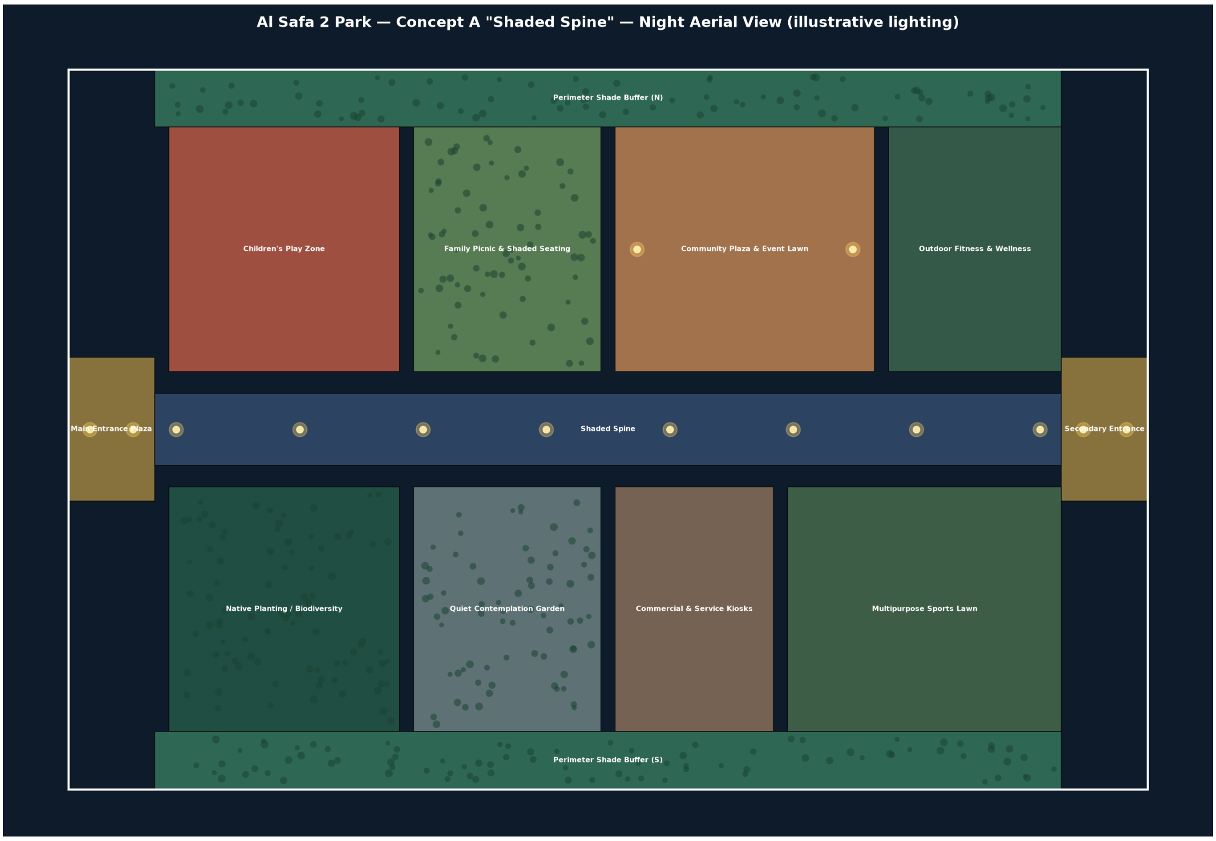


Annual O&M breakdown + 10-year total cost of ownership.

7. Visualization (Phase 9)

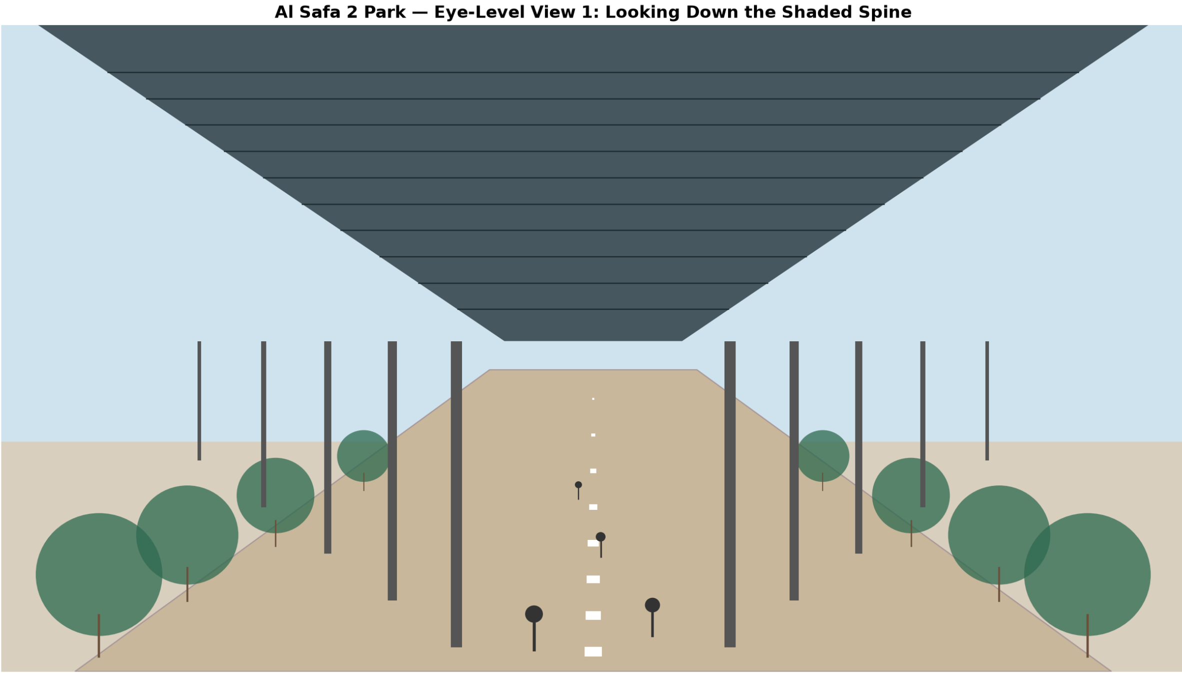


Aerial day view.



Aerial night view (illustrative lighting).

Al Safa 2 Park — Eye-Level View 1: Looking Down the Shaded Spine



Eye-level view down the Shaded Spine.

8. Why This Proposal Wins

- **Innovation (20%):** AI used for real quantitative analysis (8,760-hr solar, shade simulation, water & demand models) — not just pretty renders.
- **Human-Centered/Sustainability (20%):** directly solves the #1 real problem (heat) with proven 99.2% shaded circulation, +3 comfortable months (Heat Index), a real ~5,700 m3/yr water budget, and ~2.1 t CO2/yr sequestration.

- **AI Integration (20%):** every phase has a reproducible Python script and a 'Program Proof' code appendix — full transparency.
- **Design Quality/UX (15%):** one legible organizing idea, real personas, day/night activation.
- **Feasibility (20%):** highest-feasibility concept, costed to the AED 35M budget, with a real computed water budget.
- **Presentation (5%):** a consistent, professional, fully-sourced document set.

Integrity Statement

Every quantitative figure in this proposal is either (a) computed from real astronomy/geometry, (b) sourced from a named real dataset with a retrieval date, or (c) a clearly-labeled planning assumption. Genuine data gaps (exact DWG park boundary, on-site GIS/soil/noise) are flagged, never fabricated — because integrity is what separates a credible submission from a disqualified one.